

Rate Monotonic Scheduling (RMS) Algorithm

- Input
 - Set of n periodic tasks
 - Each task τ_i has:
 - Execution time C_i
 - Period T_i
 - Deadline $D_i = T_i$
- Algorithm Steps
 - Assign Priorities
 - Sort all tasks in ascending order of period.
 - Task with smallest period gets highest priority.
 - Initialize System Clock
 - Set time $t=0$.
 - Task Release
 - At each period T_i , release a new instance of task τ_i
 - Ready Queue Management
 - Maintain a ready queue ordered by fixed priority (shorter period $\rightarrow$ higher priority).
 - Scheduling Decision
 - At any scheduling point (task release, completion, or preemption):
 - Select the highest-priority ready task.
 - Preemption Rule
 - If a higher-priority task becomes ready, preempt the currently running task.
 - Execution
 - Run the selected task until:
 - It finishes execution, or
 - It is preempted by a higher-priority task.
 - Deadline Check
 - If any task does not complete before its next period, deadline miss occurs.

Deadline Monotonic Scheduling (DMS) Algorithm

- Input
 - Set of n periodic tasks
 - Each task τ_i has:
 - Execution time C_i
 - Period T_i
 - Relative Deadline $D_i \leq T_i$
- Algorithm Steps
 - Assign Priorities
 - Sort all tasks in ascending order of relative deadlines.
 - Task with shortest deadline gets highest priority.
 - Initialize System Clock
 - Set time $t=0$.

- Task Release
 - At each period T_i , release a new instance of task τ_i
- Ready Queue Management
 - Maintain a ready queue ordered by fixed priority (shorter deadline $\rightarrow$ higher priority).
- Scheduling Decision
 - At any scheduling point (task release, completion, or preemption):
 - Select the highest-priority ready task.
- Preemption Rule
 - If a higher-priority task becomes ready, preempt the currently running task.
- Execution
 - Run the selected task until:
 - It finishes execution, or
 - It is preempted by a higher-priority task.
- Deadline Check
 - If any task does not complete before its next period, deadline miss occurs.

Earliest Deadline First (EDF) Scheduling Algorithm

- Input
 - Set of n periodic tasks
 - Each task τ_i has:
 - Execution time C_i
 - Period T_i (for periodic task)
 - Absolute Deadline d_i
- Algorithm Steps
 - Initialize System Clock
 - Set time $t=0$.
 - Task Release
 - When a task is released, assign it an absolute deadline:
 - $d_i = \text{release time} + D_i$
 - Ready Queue Management
 - Maintain a ready queue sorted by earliest absolute deadline.
 - Scheduling Decision
 - At any scheduling point (task release, completion, or deadline update):
 - Select the task with the earliest deadline.
 - Preemption Rule
 - If a newly released task has an earlier deadline than the running task, preempt the current task.
 - Execution
 - Run the selected task until:
 - It finishes execution, or
 - It is preempted by another task with an earlier deadline.
 - Deadline Check
 - If a task does not complete before its absolute deadline, a deadline miss occurs.